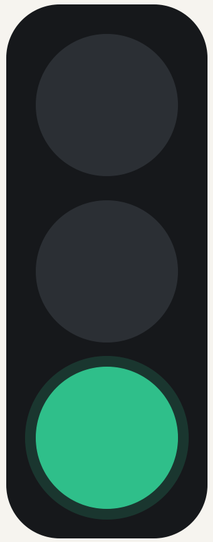
Signal Learner

A few days of watching, then a better timing plan for an engineer to approve.



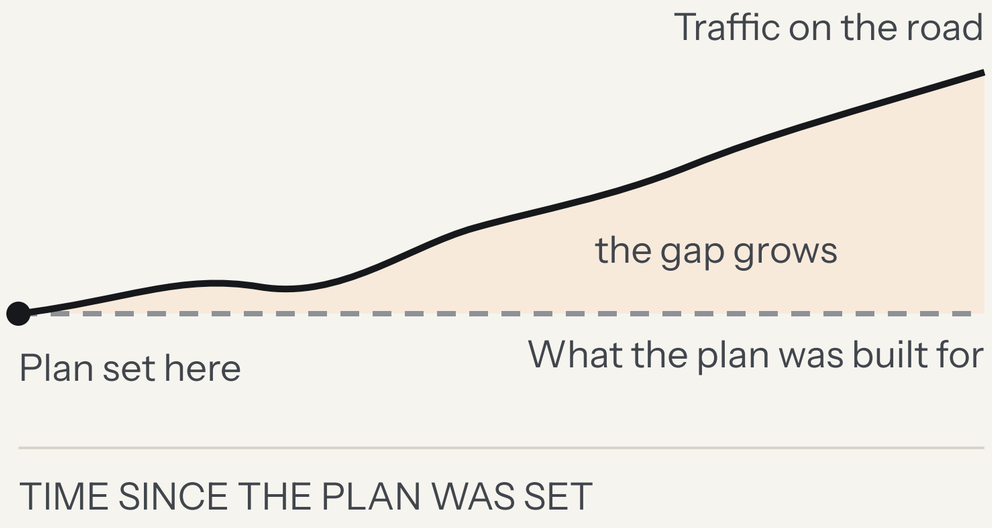
MONTA VISTA AI CLUB
MONTA VISTA HIGH SCHOOL, CUPERTINO

THE PROBLEM

Traffic changes, but a timing plan stays the same until someone retimes it.

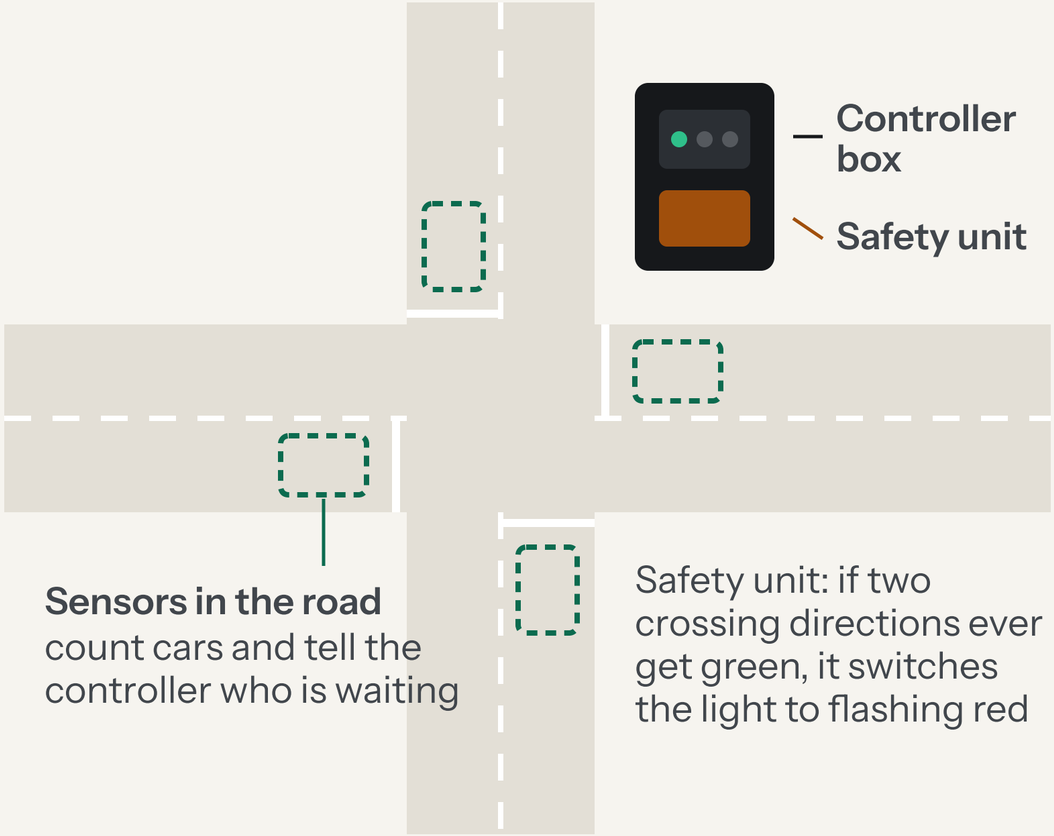
> “the traffic was backed up all the way to Glenbrook today.”
>
> SAARTH, OCT 21

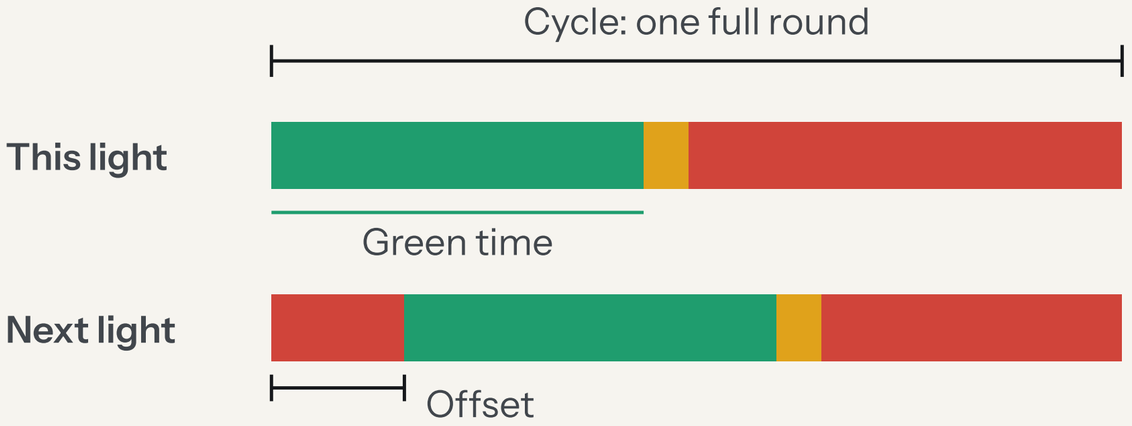
Glenbrook Apartments sit near Stevens Creek Blvd and Mary Ave. Retiming a road takes an engineer, fresh car counts and time, so plans can fall behind the traffic.



Sketch of the idea, not measured data.

A small computer at the corner follows a timing plan.





Cycle One full round of greens for every direction.

Green time How long one direction gets to go.

Offset How many seconds after its neighbor a light turns green. Neighbors share one cycle, so cars can hit greens in a row.

IS THIS ALREADY SOLVED?

Partly. Smart signal systems exist, but few signals have them.

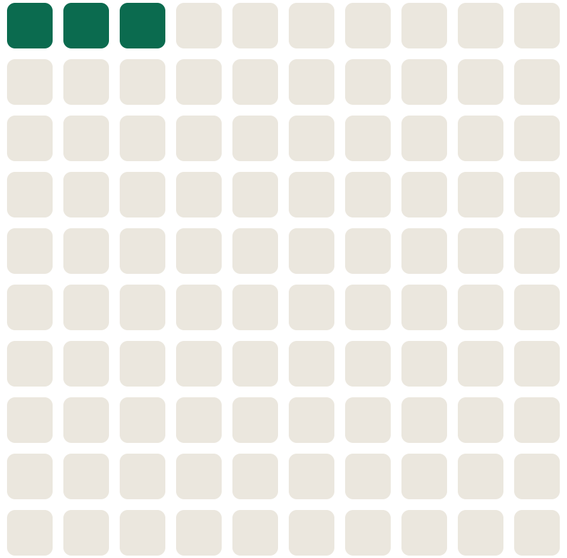
CHANGE THE TIMING ON THE GO

SCOOT · SCATS · InSync · Surtrac · SynchroGreen

RECOMMENDS, ENGINEERS DECIDE

Google Green Light

> Setting up a smart system takes money, new sensors and time.
>
> **Our spot: the cheap first step, using sensors a city already has.**
>
> We do not know yet what Cupertino runs. We asked.



About 3 in 100 signals had adaptive timing, meaning the system changes the plan by itself.

From a 2015 article. The figure is old.

OUR IDEA

Six steps, and it never touches a real light.

1 Map

Draw the road from the public map. Old timing keeps running.

2 Watch

Read the road sensors for a few days. Change nothing.

3 Learn

Learn the traffic pattern and build a practice copy of the road.

4 Practice

Try many timings on the copy, over many rush hours.

5 Recommend

Suggest one bounded change and its reason. An engineer approves or rejects.

6 Check

Keep reading the sensors. Go back to the old timing if traffic gets worse.

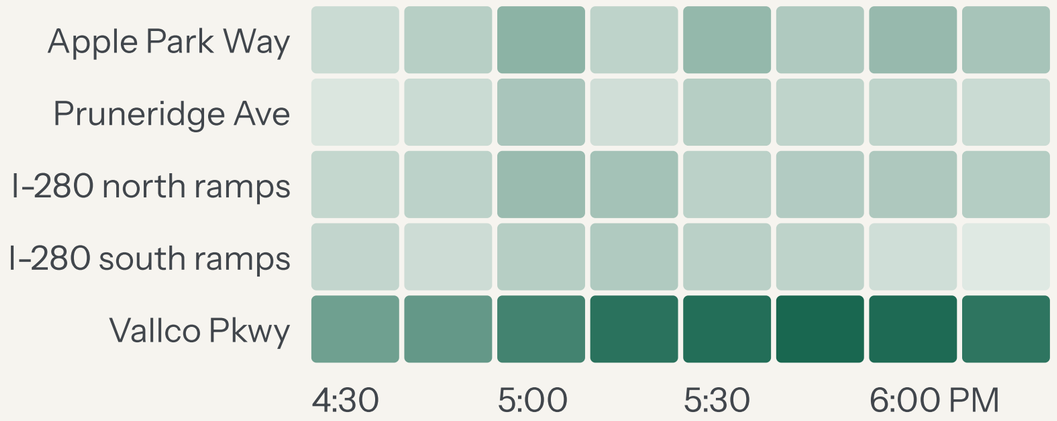
It never touches a real light. In our tests, the simulator applies an approved change.

WHERE THE AI IS

SIMULATED

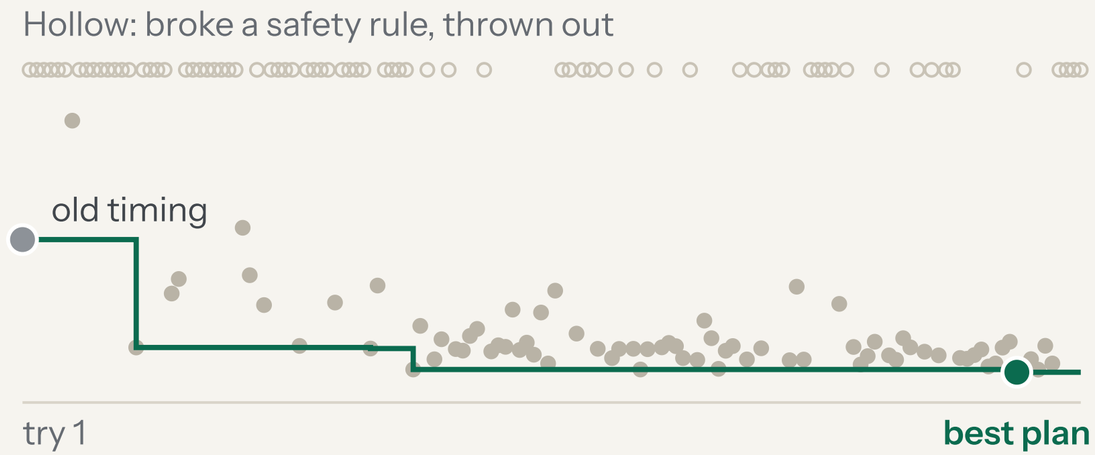
It learns the traffic pattern, then searches 150 timings on a practice copy.

1. LEARN: CARS PER 15 MINUTES, WOLFE RD



Learned from 3 watched rush hours. Darker means busier.

2. SEARCH: 150 TRIES ON THE PRACTICE COPY



Each dot is one try. Lower is better.

Model-based search (Optuna). After each try, it updates its guess of which settings look good and tries there next.

Why no neural network? Published methods needed about 100 to 1,400 simulated runs, and we found no live signal using one.

SIMULATED

One card: old and new settings, the reason, and the expected effect.

SIGNAL	CYCLE, S	OFFSET, S	MAIN GREEN ON WOLFE RD, S	REASON
Vallco Pkwy	65 → **88**	30 → **50**	22 → **38**	Learned peak: 408 cars per 15 min
I-280 south ramps	65 → **88**	0 → **58**	31 → **42**	Learned peak: 360 cars per 15 min
I-280 north ramps	65 → **88**	20 → **4**	21 → **35**	Learned peak: 502 cars per 15 min
Pruneridge Ave	65 → **88**	36 → **83**	10 → **20**	Learned peak: 344 cars per 15 min
Apple Park Way	65 → **88**	42 → **47**	17 → **34**	Learned peak: 360 cars per 15 min

Expected effect, in the practice copy: delay score 138 s to 80 s, 42% lower. Offsets move so the lights still line up on the longer cycle.

Unchanged: the order of the lights, yellow and all-red times, and minimum greens.

ENGINEER DECISION

☐ Approve ☐ Reject

HOW WE KEPT THE TEST FAIR

The learner never saw the road's true traffic, and the rules came first.

WHAT THE LEARNER GETS

- The road map
- The list of signals
- The old timing
- The sensor feed: car counts every 5 minutes



HIDDEN FROM THE LEARNER

The true traffic: every trip and every route.

A test fails if learner code ever opens it.

Rules written first. Frozen in our code history before the test ran.

Two roads it never saw. Wolfe Rd and De Anza Blvd, one try each.

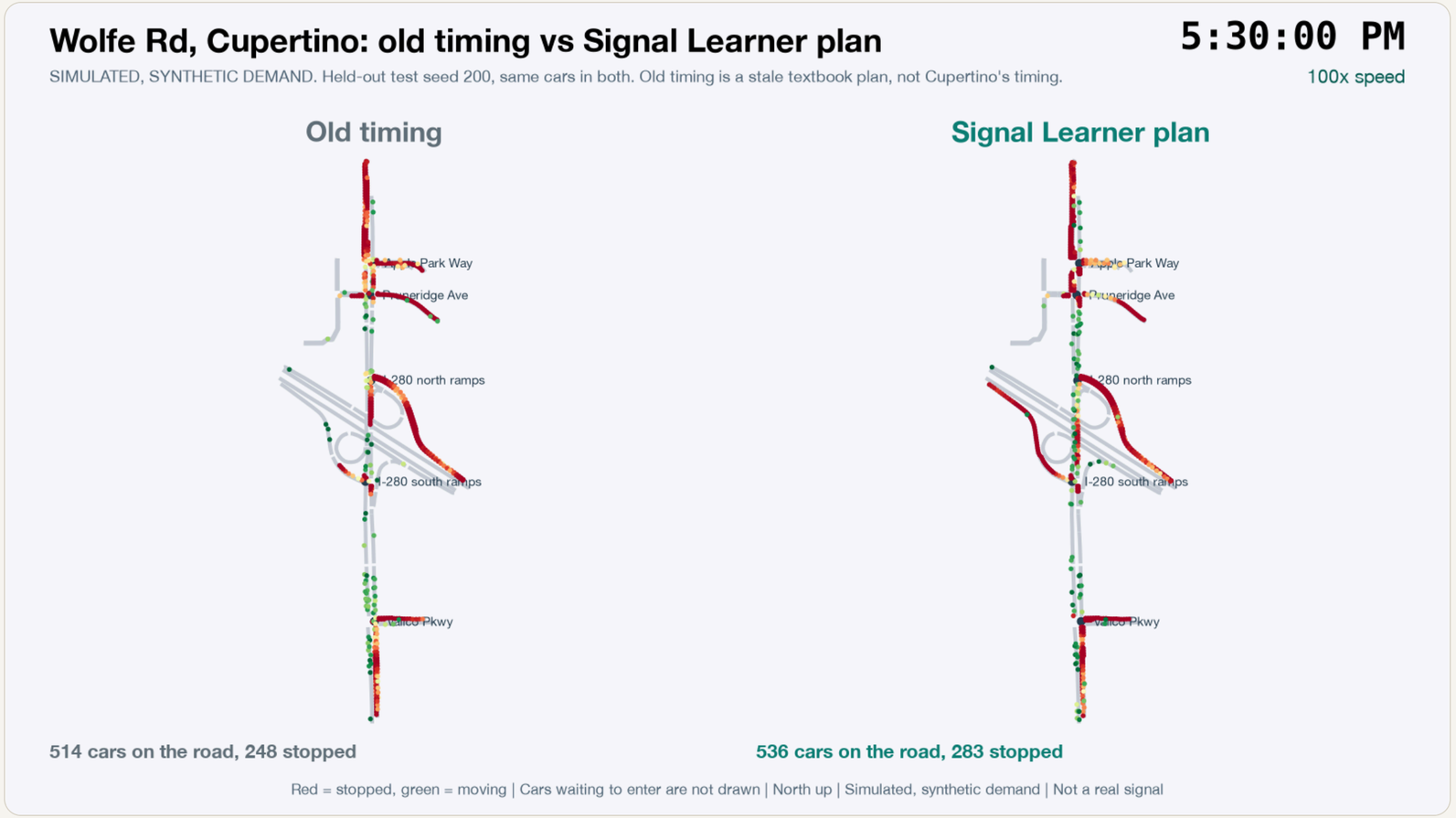
Old timing is a stand-in. Built from out-of-date traffic. Not Cupertino's timing.

New-road traffic is made up by us. We call it synthetic.

WATCH IT: WOLFE RD

SIMULATED

Same cars, same rush hour: old timing on the left, the learner's plan on the right.



SIMULATED, SYNTHETIC DEMAND

One test rush hour, played 100 times faster than real life.

Red dots are stopped cars. Green dots are moving.

Why the right side can show more cars: cars still waiting to get onto the road are not drawn. Old timing leaves more of them stuck outside.

RESULT 1: WOLFE RD, A ROAD IT NEVER SAW

SIMULATED

On Wolfe Rd, the learner's plan cut the delay score by 26%.

Old timing



870 s

Fresh retiming with perfect counts (textbook method, true traffic)



846 s −3%

Signal Learner (saw only the sensors)



643 s −26%

Delay score: average seconds lost per trip compared with an empty road, plus 900 s for every trip that never finishes or gets stuck. It is a score, not stopwatch time.

TRIPS THAT NEVER FINISHED, PER RUSH HOUR

968 to 217

Better on all 20 test rush hours.

ONE GROUP GOT WORSE

Trips from the entry point with the longest waits lost about **203 s more** each: 2,566 s to 2,769 s.

RESULT 2: DE ANZA BLVD, THE SECOND UNSEEN ROAD

● SIMULATED · POST-HOC CHECK

On De Anza Blvd it said “no change”, and a later check says that was right.

THE CARD SAID

No change.

Old timing stays. Knowing when to do nothing matters.

POST-HOC: CHECKED AFTERWARD ON 20 NEW RUNS

Old timing



275 s

The plan it turned down



292 s 6.4% worse



13 of 20 runs worse

1.45 to 12.15

stuck-car events per rush hour

The practice copy was badly wrong here. One lane is shared by straight and right-turning cars, and the copy sent 3,791 cars right. It expected 1,740 s of delay; the real road had about 270 s. A safety rule blocked the plan because it made more cars stuck.

ALL FOUR ROADS

SIMULATED

It beat old timing on three roads and chose no change on the fourth.

ROAD	USED FOR	TRAFFIC	OLD TIMING	FRESH RETIMING, PERFECT COUNTS	SIGNAL LEARNER	V5, SAW THE TRUE TRAFFIC
Homestead Rd	PRACTICE	Synthetic	130 s	113 s −13%	**77 s −41%**	not run
Stevens Creek Blvd	PRACTICE	VTA 2024 counts	682 s	661 s −3%	**471 s −31%**	**369 s** −46%
Wolfe Rd	UNSEEN TEST	Synthetic	870 s	846 s −3%	**643 s −26%**	not run
De Anza Blvd	UNSEEN TEST	Synthetic	270 s	481 s +78%	270 s no change	not run

Delay score per trip, lower is better. Our rules were tuned on the practice roads (10 test rush hours each), so only the unseen roads (20 each) are a fair test.

v5 is our earlier optimizer. It had full access to the true traffic and still wins on Stevens Creek. Stevens Creek uses VTA 2024 counts. **Credit: VTA.**

WHAT IS NOT PROVEN

What this does not prove yet.

1 **Simulation only.**
No real signal, car or street was involved.

2 **Made-up traffic on the new roads.**
Wolfe and De Anza traffic was generated by us.

3 **Old timing is a stand-in.**
We built it from out-of-date traffic. It is not Cupertino's timing.

4 **The practice copy was off.**
On Wolfe it had 24% fewer trips than the true traffic.



5 **The undo check passed once, failed once.**
Passed on Wolfe. On De Anza it undid a harmless change on 40% of runs.



6 **Two unseen roads is a small test.**
It does not show it works on any road.

HOW THIS CONNECTS TO CUPERTINO

Three steps to a real city, and the first two need no connection to the lights.

STEP 1

Read

Use the city's data exports or controller logs. ATSPM, a free open tool for signal logs, can turn them into our sensor feed.

STEP 2

Recommend

An engineer types the approved change into the city's system by hand, the way Google Green Light works. No connection needed.

STEP 3, MUCH LATER

Automatic

Send changes straight to the lights using NTCIP 1202, the standard language signal controllers speak. Only with city approval.

WHERE WE ARE

We emailed the city's Traffic and Engineering and Signal Maintenance teams. **No answer yet.**

Cupertino selected a Cubic Trafficware central system, its software for running lights from one office, in 2019. We do not know how it is set up today.

HACKATHON WEEK

What we fix and test next.



Fix the shared-lane error

Teach the copy to split straight and right-turning cars that share one lane, the error that broke De Anza.



Watch longer

More than 3 rush hours, so the undo check learns what a normal jam day looks like.



Test on fresh roads

New roads, picked and written down before any run, one try each.

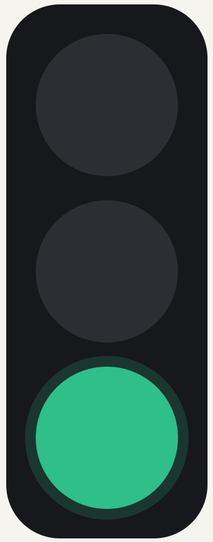


Use real city data

If Cupertino sends data, run the same six steps on it.

In simulation, Signal Learner found a better plan on one road it never saw, and knew to change nothing on another.

Help us test it on *real data.*



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